

ASTRAEDU – AI STUDY ASSISTANT

Capstone Project Report

Sustainable Development Goal (SDG)

SDG 4: Quality Education

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1. Executive Summary

AstraEdu is an AI-powered Study Assistant chatbot developed using Botpress to support Sustainable Development Goal 4 (Quality Education). The chatbot helps students improve their academic performance through study planning, exam preparation, goal setting, time management, motivation, and learning guidance. By providing instant educational assistance through conversational AI, AstraEdu aims to make learning more accessible, organized, and engaging for students.

2. SDG Selection

Selected SDG

SDG 4 – Quality Education

Why This SDG?

Quality education is essential for personal development, career growth, and societal progress. Many students face challenges related to study planning, productivity, motivation, and academic guidance. AstraEdu addresses these challenges by providing AI-powered educational support and helping learners develop effective study habits.

3. Problem Statement

Many students struggle with:

- Creating effective study plans
- Managing their time efficiently
- Preparing for examinations
- Staying motivated throughout their learning journey
- Setting realistic academic goals

Without proper guidance and support, students may experience reduced productivity and lower academic performance.

4. Project Objective

The primary objective of AstraEdu is to develop an AI-powered educational assistant capable of:

- Providing study planning assistance
 - Supporting exam preparation
 - Encouraging effective time management
 - Assisting with goal setting
 - Delivering motivational support
 - Promoting productive learning habits
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5. Proposed Solution

AstraEdu is a conversational AI chatbot developed using Botpress.

The chatbot uses Natural Language Processing (NLP) and Conversational AI techniques to understand user queries and provide educational guidance.

Students can interact with AstraEdu to receive:

- Study schedules
 - Exam preparation strategies
 - Time management recommendations
 - Goal-setting guidance
 - Motivational support
 - Learning improvement techniques
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6. System Features

6.1 Study Planning

- Daily study schedules
- Structured learning routines
- Study habit recommendations

6.2 Exam Preparation

- Revision strategies
- Preparation techniques
- Practice recommendations

6.3 Time Management

- Productivity techniques
- Task prioritization
- Pomodoro recommendations

6.4 Goal Setting

- SMART goal framework
- Milestone planning
- Progress tracking suggestions

6.5 Motivation

- Inspirational quotes
- Positive reinforcement
- Student encouragement

6.6 Learning Tips

- Active learning strategies
- Note-taking methods
- Memory improvement techniques

6.7 Fallback Response

- Handles unknown inputs gracefully
 - Guides users back to supported features
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7. Technology Stack

Component	Technology Used
Chatbot Platform	Botpress Cloud
AI Technology	Conversational AI
NLP	Natural Language Processing
Documentation	Microsoft Word
Repository	GitHub
Deployment	Botpress Cloud

8. System Workflow

Workflow Diagram

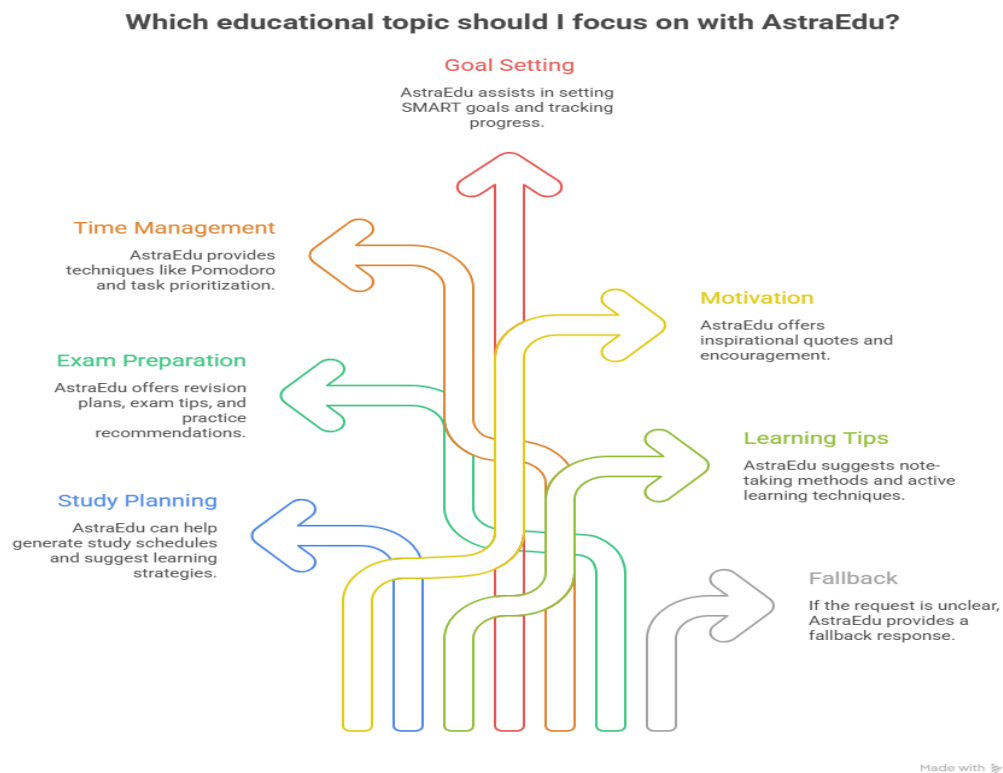


Figure 1: AstraEdu System Workflow

Workflow Description

1. User initiates conversation.
 2. AstraEdu displays a welcome message.
 3. User selects an educational topic.
 4. AI processes the request.
 5. Appropriate educational guidance is generated.
 6. Unknown inputs trigger fallback responses.
 7. User may continue interaction or end the session.
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9. Implementation

Development Process

Step 1

Created AstraEdu chatbot using Botpress.

Step 2

Configured chatbot personality and educational objectives.

Step 3

Implemented educational conversation flows.

Step 4

Tested all chatbot interactions.

Step 5

Published chatbot and generated deployment link.

Step 6

Exported Botpress project and uploaded project assets to GitHub.

10. Project Screenshots

Figure 2: Workflow Design

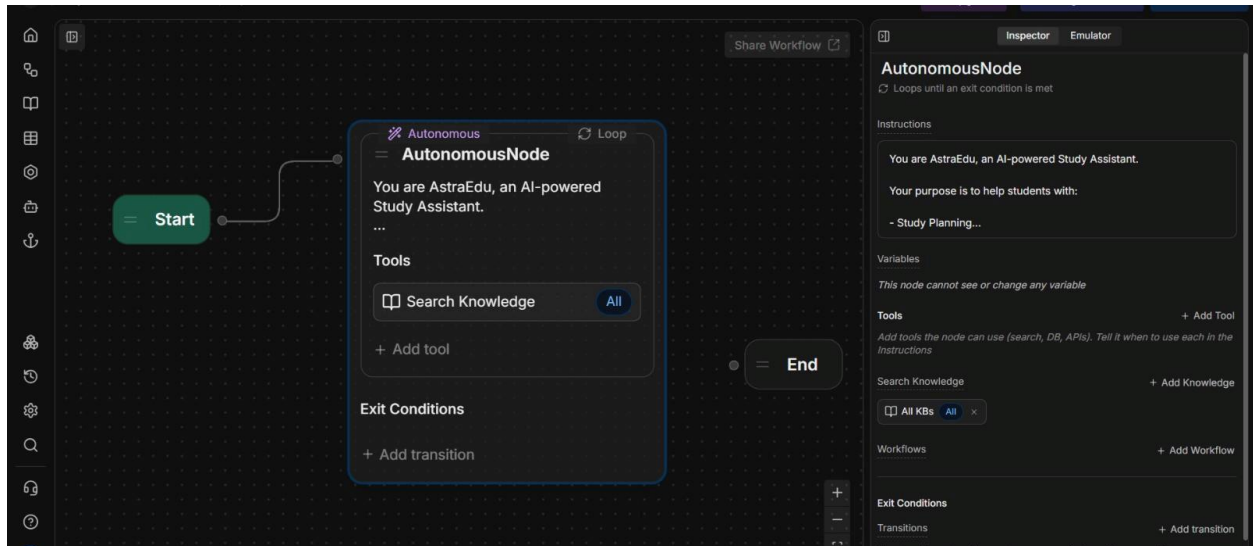


Figure 3: Welcome Message

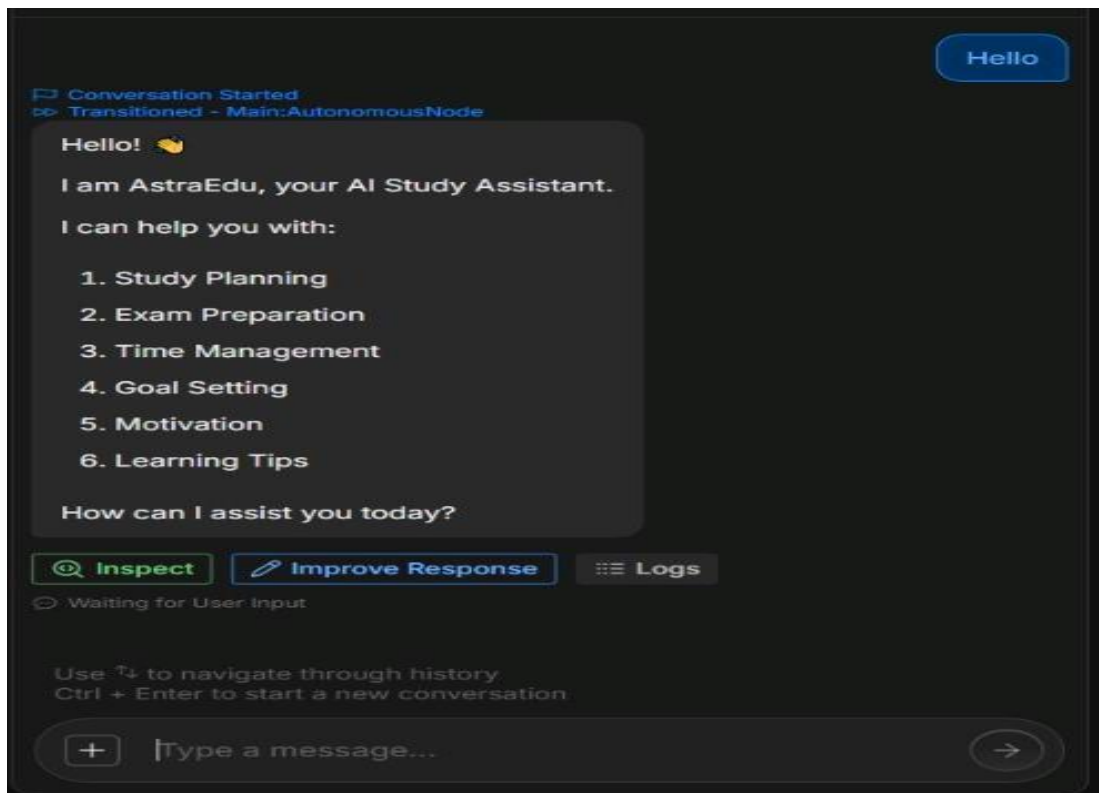


Figure 4: Study Planning Interaction

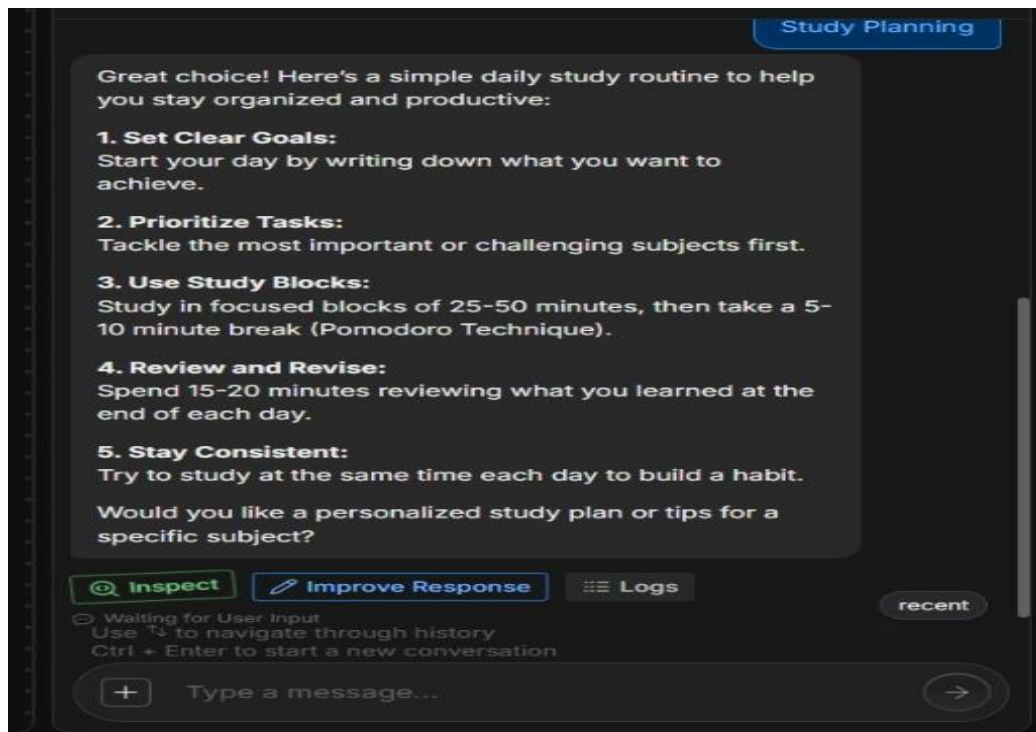


Figure 5: Exam Preparation Interaction

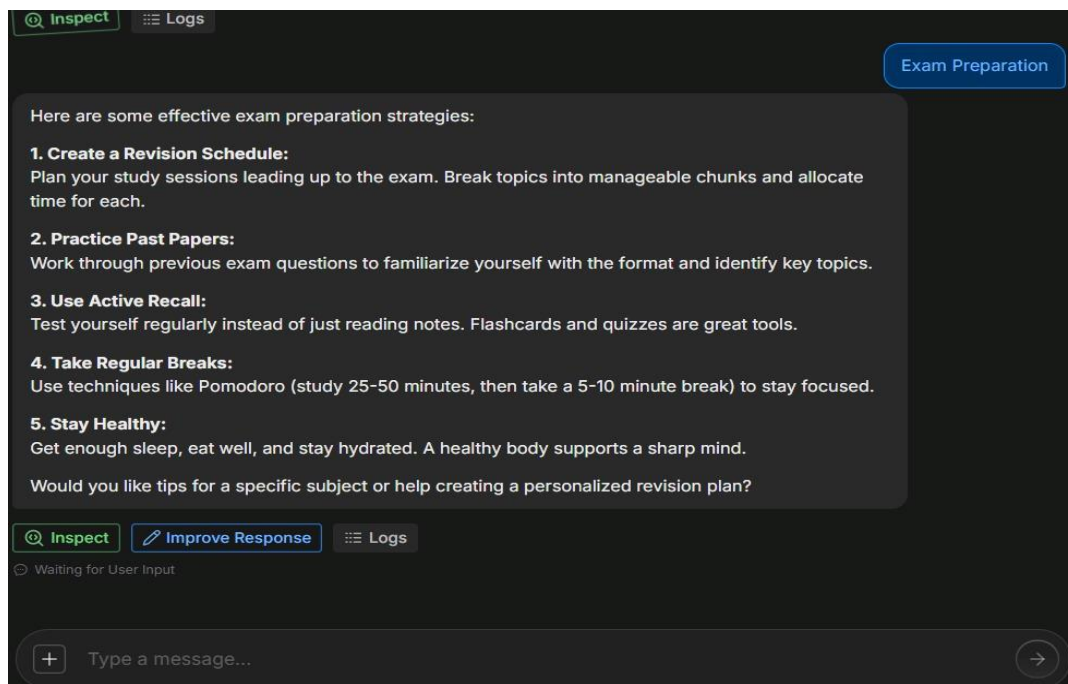


Figure 6: Time Management Interaction

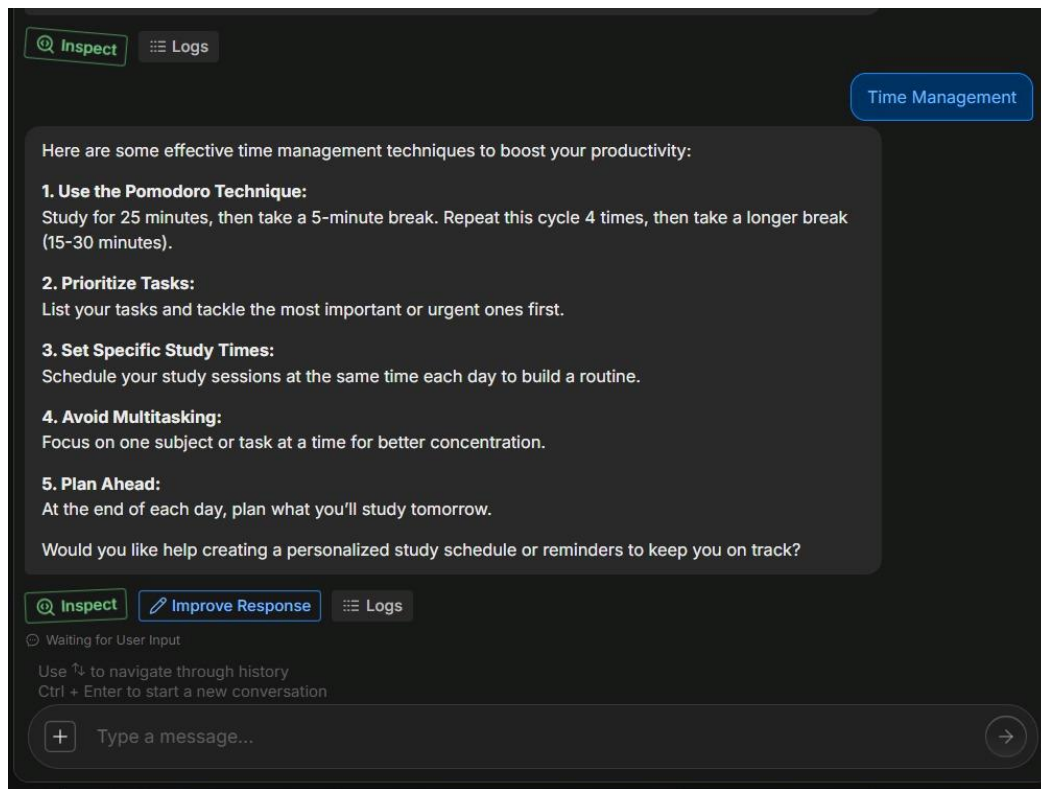


Figure 7: Goal Setting Interaction

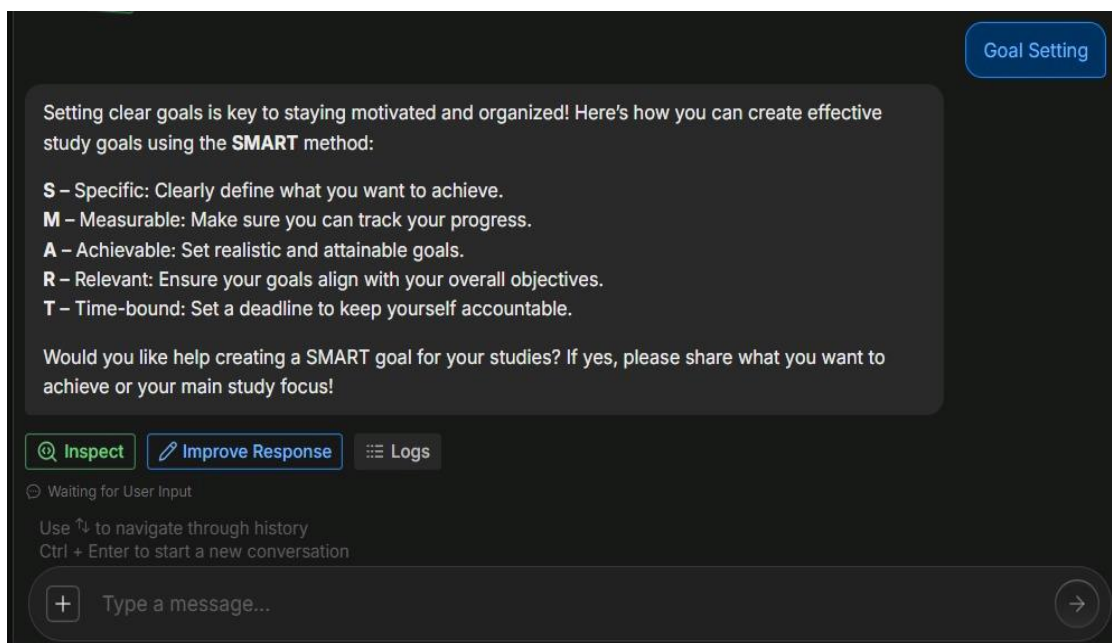


Figure 8: Learning Tips Interaction

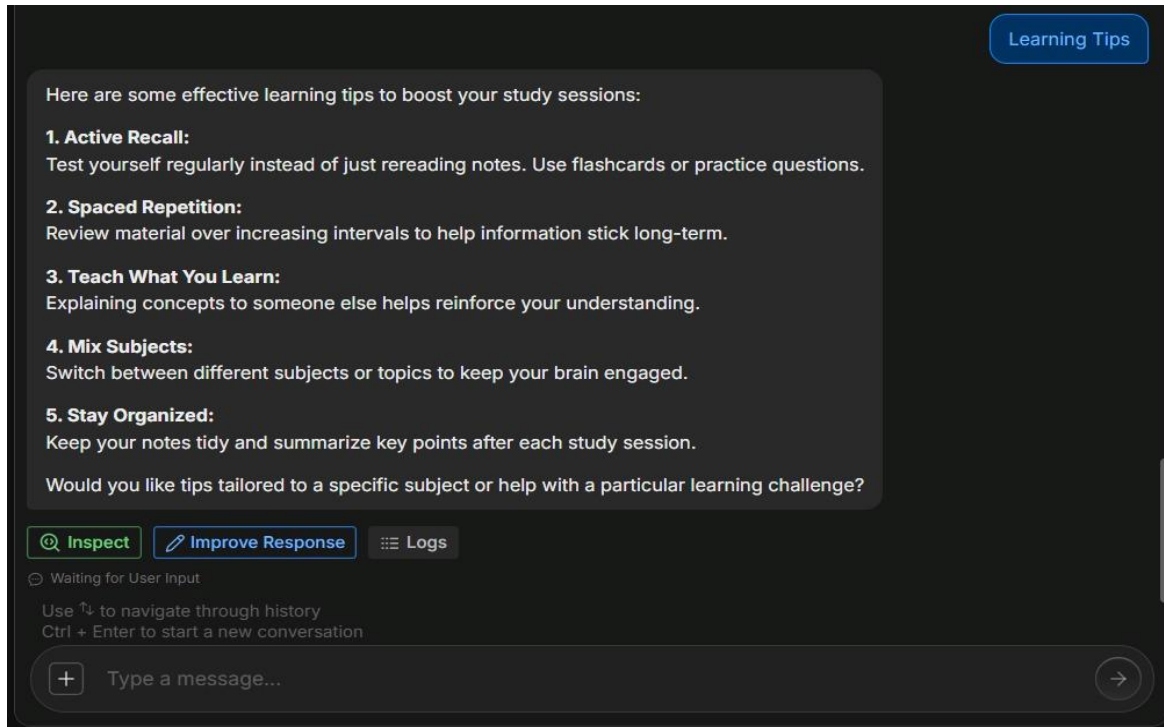


Figure 9: Fallback Response

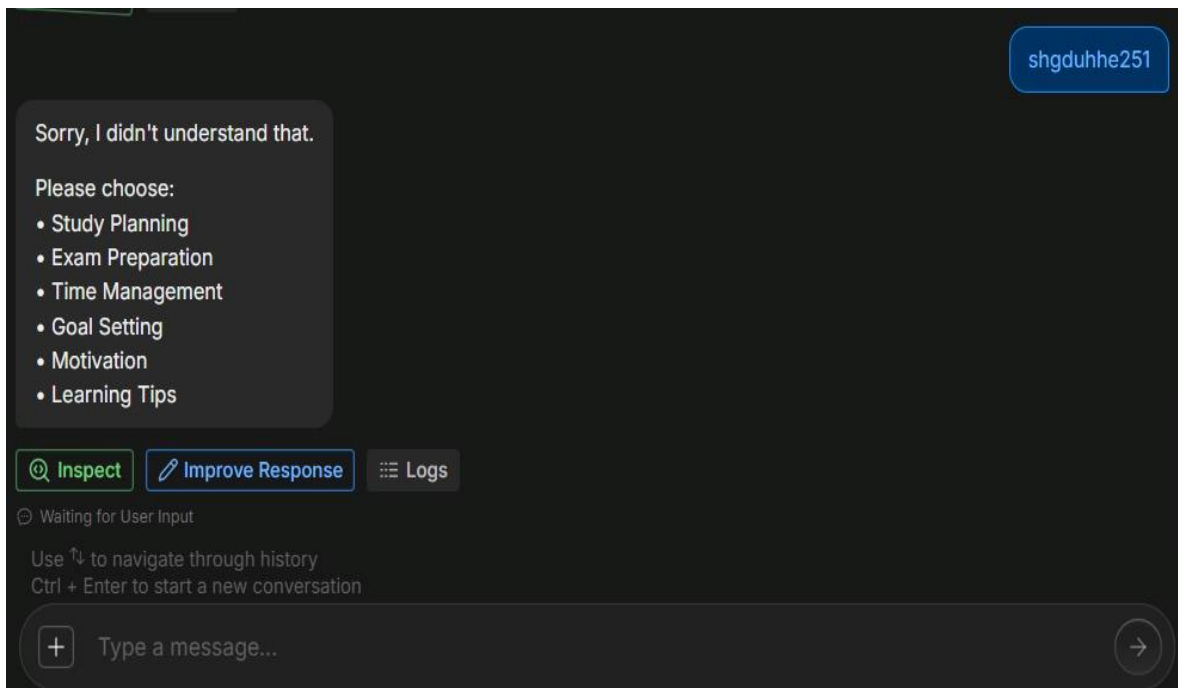
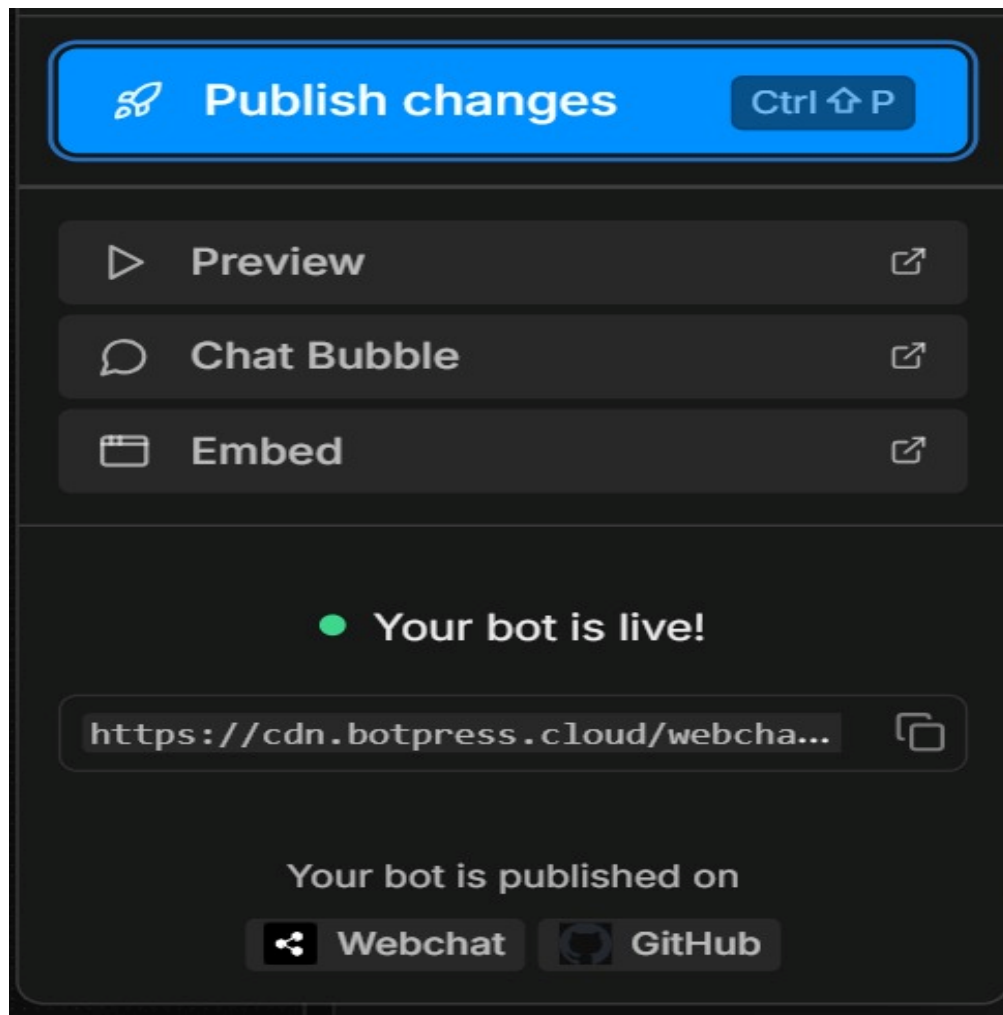


Figure 10: Published Chatbot



11. GitHub Repository

Repository Link

<https://github.com/Omkar-Gujar/AstraEdu-AI-Study-Assistant>

12. Bot Deployment Link

Public Chatbot Link

<https://cdn.botpress.cloud/webchat/v3.6/shareable.html?configUrl=https://files.bpcontent.cloud/2026/06/10/13/20260610130044-D0C1TFKP.json>

13. Future Scope

Future improvements may include:

- Voice-enabled interactions
 - Multi-language support
 - Personalized learning recommendations
 - Learning analytics dashboard
 - Integration with educational platforms
 - Mobile application deployment
 - AI-powered progress tracking
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14. Conclusion

AstraEdu successfully demonstrates the use of Artificial Intelligence in supporting Sustainable Development Goal 4 (Quality Education). The chatbot provides accessible educational guidance through conversational AI, helping students improve study planning, exam preparation, time management, motivation, and overall productivity. This project showcases the practical application of AI technologies in addressing educational challenges and promoting effective learning experiences.

References

1. Botpress Documentation
2. United Nations Sustainable Development Goals (SDG 4)
3. GitHub Documentation
4. Conversational AI and NLP Learning Resources